MILITARY SPECIFICATION

DETONATOR, DELAY, ELECTRIC: PA506; PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, and the loading, assembling and packing for one type of electric delay detonator designated Detonator, Delay, Electric: PA506.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of
MIL-B-117 - Bag, Sleeve and Tubing - Interior Packing
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FEDERAL

TT-I-1795 - Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-650 - Explosive: Sampling, Inspection and Testing

DRAWINGS

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
9311158 - Detonator, Delay Electric, PA506

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)").

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing (dwg.) C9311158, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content of barium chromate/ boron. - The moisture content of barium chromate/boron at the time and place of loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.2.

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3.4 Moisture content of lead azide. - The moisture content of lead azide at the time and place of loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.3.

3.5 Moisture content of HMX. - The moisture content of the HMX at the time and place of loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.4.

3.6 Functioning.

3.6.1 Output. - The detonator shall fire on the first attempt, and shall make a minimum depression of 0.0063 inches in the steel witness plate when tested as specified in 4.5.9.1.

3.6.2 Waterproofness. - The detonator shall comply with the output requirements of 3.6.1 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 70 degrees F. plus or minus 5 degrees, when tested as specified in 4.5.9.2.

3.6.3 Time. - The detonator firing time shall comply with the requirements given on Dwg. C9311158 when tested as specified in 4.5.9.3.

3.7 Environmental

3.7.1 Resistance between electrode and cup, thermal shock and vibration. - The detonator shall comply with the resistance requirement of Dwg. C9311158, and with the output requirements of 3.6.1 after being exposed to thermal shock and vibration as specified in 4.5.10.1.

3.7.2 Cold temperature. - The detonator assembly shall comply with the output requirements of 3.6.1 after being conditioned at minus 65 degrees F plus or minus 5 degrees for two hours and tested as specified in 4.5.10.2.

3.7.3 Hot temperature. - The detonator assembly shall comply with the output requirements of 3.6.1 after being conditioned at plus 160 degrees F plus or minus 5 degrees for two hours and tested as specified in 4.5.10.3.

3.8 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for submission of first article samples shall be specified in the contract.

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3.9 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free from burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Bridge Assembly	9245693	50
Plug Assembly	9245694	50
Electrode	9245695	50
Header	9245696	50
Pellet	9245698	50
Detonator, Delay Electric, PA506	9311158	325
Bridged Plug and Ignition Assembly	9311159	50
Cup	9311160	50

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	PA506 Delay Electric Detonator and Components				See Below
					NEXT HIGHER ASSEMBLY
					-
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
	Cup				
	(Dwg. 9311160)				
	Examination for defects	50		3.2	4.4.2.1
	Electrode				
	(Dwg. P-9245695)				
	Examination for defects	50		3.2	4.4.2.2
	Header				
	(Dwg. P-9245696)				
	Examination for defects	50		3.2	4.4.2.3
	Pellet				
	(Dwg. P-9245698)				
	Examination for defects	50		3.2	4.4.2.4
	Plug Assembly				
	(Dwg. P-9245694)				
	Examination for defects	50		3.2	4.4.2.5
	Pull test of plug assembly	50		3.2	4.5.6/(a)
	Insulation resistance between electrode and header	50		3.2	4.5.7
	Bridged Plug and Ignition Assembly				
	(Dwg. 9311159)				
	Examination for defects	50		3.2	4.4.2.6

NOTES:

(a) Reject if one or more is defective

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TABLE I - FIRST ARTICLE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	PA506 Delay Electric Detonator and Components				See Below
					NEXT HIGHER ASSEMBLY —
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
	Detonator, Delay Electric, PA506 (Dwg. 9311158)				
	Examination for defects	(a)		3.2	4.4.2.7
	Functioning - Output	200		3.6.1	4.5.9/4.5.9.1
	Functioning - Waterproofness	50		3.6.2	4.5.9/4.5.9.1/4.5.9.2
	Functioning - Time	(a)		3.6.3	4.5.9./4.5.9.1/4.5.9.3
	Resistance of lead circuit, thermal shock and vibration	25		3.7.1	4.5.10.1
	Environmental-cold temperature	25		3.7.2	4.5.10.2
	Environmental-hot temperature	25		3.7.3	4.5.10.3

NOTES:
(a) The sample shall consist of the 325 detonators to be functioned.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer, in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant. The second lead azide lot shall not vary from the first lot by more than three serial lot numbers.

c. HMX from not more than one lot.

d. Barium Chromate/Boron from not more than one lot.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63197(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	9311160
			NEXT HIGHER ASSEMBLY 9311161

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major					
101	Thickness of bottom		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Concentricity		0.40%	3.2	Gage
104	Contour of bottom radius		0.40%	3.2	Gage
105	Inside diameter		0.40%	3.2	Gage
106	Length		0.40%	3.2	Gage
107	Dent or distortion		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63197(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Electrode		P-9245695
			NEXT HIGHER ASSEMBLY P-9245694

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Length		0.40%	3.2	Gage
103	End finish improper		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63197(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Header		P-9245696
			NEXT HIGHER ASSEMBLY
			P-9245694

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Diameter large cavity		0.40%	3.2	Gage
103	Diameter of bottom hole		0.40%	3.2	Gage
104	Length		0.40%	3.2	Gage
105	Bottom thickness, Min		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Pellet				P-9245698
					NEXT HIGHER ASSEMBLY P-9245694
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Diameter of hole		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63197(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Plug Assembly		P-9245694
			NEXT HIGHER ASSEMBLY P-9245693

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Insulation resistance between electrode and header		100%	3.2	4.5.7
102	Gold plating improper or missing		0.40%	3.2	4.5.5
103	Pull test		0.40%	3.2	4.5.6
104	Outside diameter, max		0.40%	3.2	Gage
105	Length of electrode below bottom of header		0.40%	3.2	Gage
106	Glass pellet cracked		0.40%	3.2	Visual
Minor					
201	Thickness of header		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63197(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Bridged Plug and Ignition Assembly		9311159
			NEXT HIGHER ASSEMBLY 9311158

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spot charge contains cracks, voids or separations		0.40%	3.2	Visual
102	Spot charge missing		0.40%	3.2	Visual
103	LMNR composition missing or does not cover spot charge completely		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63197(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Detonator, Delay, Electric: PA506	1 OF 1	9311158
			NEXT HIGHER ASSEMBLY: –

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
Critical					
1	Cup cracked or split		100.0%	3.2	Visual
Major					
101	Resistance between electrode and cup		100.0%	3.2	4.5.8
102	Length of electrode		0.40%	3.2	Gage
103	Length, crimped end to base		0.40%	3.2	Gage
104	Depth to spherical radius		0.40%	3.2	Gage
105	Diameter of cup		0.40%	3.2	Gage
106	Spherical radius missing or incorrect		0.40%	3.2	Gage
107	Crimp not full 360 degrees		0.40%	3.2	Visual
108	Sealant missing		0.40%	3.2	Visual
109	Functioning - Output	200		3.6.1	4.5.9.1
110	Functioning - Waterproofness	50		3.6.2	4.5.9.2
111	Functioning - Time	(a)		3.6.3	4.5.9.3
112	Environmental-cold temperature	25		3.7.2	4.5.10.2
113	Environmental-hot temperature	25		3.7.3	4.5.10.3
114	Resistance between electrode and cup, thermal shock and vibration	25		3.7.1	4.5.10.1
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

(a) The sample will consist of detonators fired for Major Defect Test 109.

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test methods and procedures.

4.5.1 Materials, components and processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.2 Moisture content of barium chromate/boron. The moisture content of barium chromate/boron shall be determined for each lot at the time of loading. A sample in sufficient quantity to perform the test shall be selected and tested as specified herein. The moisture content of the barium chromate/boron shall be determined in accordance with MIL-STD-650, Method 101.5.

4.5.3 Moisture content of lead azide. - The moisture content of lead azide shall be determined as specified in 4.5.2 except, a two gram sample shall be selected and the dish covered with a sheet of polytetrafluoroethylene.

4.5.4 Moisture content of HMX. - The moisture content of HMX shall be determined as specified in 4.5.3.

4.5.5 Gold plating electrodes. - The gold plating shall be verified as specified in MIL-G-45204. (test samples shall not be returned to the lot).

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4.5.6 Pull test of electrode of plug assembly. - The plug assembly shall be assembled to the equipment and tested in accordance with Method 211 of MIL-STD-202. The electrode and header shall be observed for compliance with the drawing requirement (test samples shall not be returned to the lot).

4.5.7 Insulation resistance between electrode and header of plug assembly. - The required D.C. voltage shall be applied to the plug assembly between the electrode and the header for not less than one or more than two seconds. If the resistance measures less than the specified resistance, the plug assembly shall be classed defective and removed from the lot.

4.5.8 Resistance between electrode and cup of detonator. The resistance between the electrode and the cup of the detonator after assembly shall be measured with an ohmmeter which limits the current to one (1) milliampere. If any detonator fails to comply with the requirements specified on the applicable drawing, it shall be removed from the lot.

4.5.9 Functioning. - The detonator assemblies shall be temperature-conditioned at 70 plus or minus 5 degrees F for two (2) hours, minimum, prior to performing the functioning tests. The tests shall be performed as specified in MIL-STD-331, Test 301.1, except that the thickness of the witness plate shall be 1/8 inch, minimum, and optionally round, 1.22 - .01 dia. No confining sleeve shall be used. Witness plate hardness shall be Rockwell B80 to B95. The base of the detonator shall be placed directly against the center of the witness plate without tape or grease and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to each electrode. The measurement of the depression in the witness plate shall be accomplished with a dial indicator equipped with a probe having a .003 plus and minus .001 spherical radius.

4.5.9.1 Output. - Two hundred (200) detonators from each lot shall be subjected to this test. The detonators shall be assembled into a test fixture and fired by applying the energy from a 100 microfarad, maximum, capacitor charged to 1.6 volts, maximum. The depression in the witness block shall be measured to determine compliance with the requirements.

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Each functioning time shall be recorded. If any detonator fails to meet the output requirements on the first attempt it shall be classed defective. If two (2) or more defectives occur, the lot shall be rejected.

4.5.9.2 Waterproofness. - Fifty (50) detonators from each lot shall be subjected to this test. The detonators shall be immersed in water to a minimum depth of two (2) inches for 48 hours, minimum. The required water temperature of 70° ± 5°F shall be maintained for the immersion period. At the end of the immersion time, the detonators shall be removed from the water, assembled into a test fixture and fired by applying the energy from a 100 microfarad, maximum, capacitor charged to 1.6 volts, maximum. The depression in the witness block shall be measured to determine compliance with the requirements. Functioning times shall be recorded. If any detonator fails to meet the requirements on the first attempt it shall be classed defective. If one or more defectives occur, the lot shall be rejected.

4.5.9.3 Time. - The functioning times of the two hundred (200) detonators subjected to the test in 4.5.9.1 shall be determined for compliance with specified drawing requirements. If two (2) or more detonators fail to comply with the functioning time requirements specified on the applicable drawing, the lot shall be rejected.

4.5.10 Environmental tests.

4.5.10.1 Resistance between electrods and cup, thermal shock and vibration. - Twenty-five (25) detonators from each lot shall be subjected to this test. The detonators shall be tested in the following sequence;

a. Measure the resistance between the electrode and the cup in accordance with 4.5.8.

b. Condition for two (2) hours, minimum, at minus 65 plus or minus 5 degrees F.

c. Remove from cold chamber and immediately transfer to the hot chamber.

d. Condition for two (2) hours, minimum, at plus 160 plus or minus 5 degrees F.

e. Remove from hot chamber and immediately transfer to the vibrator.

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f. Vibrate the detonators for 15 minutes, minimum, in a plane which is parallel to the longitudinal axis of the detonator. Sweep the frequency range of 25-500-25 hertz with two plus or minus 0.2g peak amplitude.

g. Measure resistance as in a.

h. The detonators shall be functioned as specified in 4.5.9.1 and meet the output requirements of 3.6.1.

If any detonator fails to meet the output requirement on the first attempt, it shall be classed defective. If one or more defectives occur, the lot shall be rejected.

4.5.10.2 Cold temperature. - Twenty-five (25) detonators from each lot shall be subjected to this test. The detonators shall be conditioned at minus 65 plus or minus 5 degrees F for two (2) hours, minimum. The detonators, maintained at the conditioning temperature, shall be functioned as specified in 4.5.9.1. If any detonator fails to meet the output requirement on the first attempt, it shall be classed defective. If one or more defectives occur, the lot shall be rejected.

4.5.10.3 Hot temperature. - Twenty-five (25) detonators from each lot shall be subjected to this test. The detonators shall be conditioned at plus 160 plus or minus 5 degrees F for two (2) hours, minimum. The detonators, maintained at the conditioning temperature, shall be functioned as specified in 4.5.9.1. If any detonator fails to meet the output requirement on the first attempt, it shall be classed defective. if one or more defectives occur, the lot shall be rejected.

4.5.11 Heat seal test of sealed bag. - The heat seal shall be verified as specified in MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging .

5.1.1 Level B. - Electric detonators shall be packaged in accordance with Code of Federal Regulations Title 49, Para 0-190. In addition, each detonator shall be electrically shorted during shipment and storage by means of a metallic connection between the electrode and cup. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of the assembly into the end item, the following shall be included:

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Each intermediate carton, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of MIL-B-117. The bag shall be closed. All excess air shall be exhausted from within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of mIL-P-116.

5.1.2 Level C. - Unless otherwise specified, the electric detonators shall be packaged in accordance with 5.1.1, except that the barrier bag is not required.

5.2 Packing.

5.2.1 Level B. - A suitable number of detonators, packaged as specified in 5.1.1, shall be packed in an outer box in accordance with Code of Federal Regulations Title 49: Transportation of Explosives and Other Dangerous Articles, Paragraph 173.66.

5.2.2 Level C. - Unless otherwise specified, electric detonators shall be packed as specified in 5.2.1.

5.3 Marking. - Each inner container and intermediate carton set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. (Number) Detonator, Delay, Electric: PA506
Lot NO. Date Packed (Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to TT-I-1795.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Dwg. 8796522. The DOT designation shall be: Electric Blasting Caps - Handle Carefully (see NOTE).

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e. NOTE: "The number of caps packed in one (1) outer pack shall be shown in the marking."

6. NOTES

6.1 Ordering data. - See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs, as required, to Commander, USAARADCOM, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.3 Check test for possible deterioration. - The following test is designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration. Two hundred (200) detonators shall be selected from each lot for this test.

6.3.1 Functioning. - The detonators shall be selected at random from each lot. The detonators shall be tested in the following sequence:

a. Measure the resistance between electrode and cup as specified in 4.5.8. If two (2) or more detonators fail to comply with the resistance requirement specified in Dwg 93111 the lot shall be rejected.

b. The detonators that are not classed defective in step "a" shall be functioned as specified in 4.5.9.1 and meet the output requirements of 3.6.1. If any detonator fails to meet the output requirements, it shall be classed defective. If three or more defectives occur, the lot shall be rejected.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1375-A-203

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